

PROJECT AND TEAM INFORMATION

Project Title

(Try to choose a catchy title. Max 20 words).

CampusNavX: An Intelligent Graph-Based Campus Navigation System for Smart Indoor, Outdoor, Accessible, and Emergency Route Planning Using Advanced Data Structures

Student / Team Information

<i>Team Name</i> <i>Team #</i>	<i>The Pathfinders</i> <i>DSCPP-III-2026-T377</i>
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PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

(Describe the problem and why it is important. Max 300 words).

Navigating a large university campus can be challenging, especially when students and visitors need to locate specific buildings, classrooms, laboratories, offices, or facilities. A conventional campus map may provide basic location information but may not effectively support route planning, indoor navigation, accessibility requirements, or changing campus conditions.

The proposed Campus Navigation System aims to solve this problem by providing an interactive system that enables users to select a source and destination and find suitable routes across the campus. The system will support both outdoor and indoor navigation, allowing users to navigate between buildings, floors, rooms, laboratories, and facilities. It will use graph-based representations of campus locations and pathways to calculate routes using BFS, DFS, and Dijkstra's algorithm.

The system is also designed to address situations where normal navigation is insufficient. Users will be able to find accessible routes that avoid stairs and prefer elevators or ramps. An emergency navigation mode will identify nearby emergency exits and calculate safe or shortest available evacuation routes while considering blocked pathways.

This project is important because campus environments are dynamic. Buildings and pathways may be added, removed, modified, or temporarily blocked. Therefore, the system will support dynamic campus updates and alternative route generation. It will also maintain route history and provide clear graphical visualization of routes and algorithm operations.

Beyond solving a practical navigation problem, the project demonstrates the meaningful application of data structures and algorithms in C++, including graphs, queues, priority queues/heaps, linked lists, and stacks, making it suitable as a Data Structures and Algorithms PBL project. This is based directly on the functionalities specified in your uploaded PDF, including indoor/outdoor navigation, accessibility, emergency routing, dynamic updates, and DSA implementation.

State of the Art / Current Solution (1 pt)

(Describe how the problem is solved today. Max 200 words).

Currently, campus navigation can be handled through basic campus maps or manually available location information. These approaches may help users identify buildings and general pathways, but they do not provide an integrated solution for detailed indoor and outdoor navigation.

The existing approach does not comprehensively support navigation to specific rooms, laboratories, offices, or facilities across different floors. It also lacks dedicated support for accessibility, such as avoiding stairs and prioritizing elevators or ramps.

Another limitation is handling dynamic campus conditions. When pathways are blocked or locations and pathways change, users need updated information to determine suitable alternative routes. Emergency situations also require identifying the nearest emergency exit and finding a safe alternative route, which is not addressed by a basic map.

The proposed system improves upon these limitations by representing the campus as a graph and applying algorithms such as BFS, DFS, and Dijkstra's algorithm for route calculation. It also introduces interactive visualization, accessible routing, emergency navigation, route history, dynamic pathway updates, and indoor navigation. These features provide a more comprehensive and algorithm-driven approach to campus navigation.

Project Goals and Milestones (2 pts)

(Describe the project goals and milestones. Max 300 words).

The main goal of the project is to develop an interactive Campus Navigation System in C++ that uses data structures and algorithms to provide efficient indoor and outdoor navigation across a university campus. The system should have a substantial graphical user interface and demonstrate the practical application of data structures and algorithms. The project will aim to:

1. Represent the campus as a graph containing buildings, pathways, floors, rooms, laboratories, offices, and facilities. Implement BFS, DFS, and Dijkstra's algorithm for campus traversal and shortest-path navigation.
2. Provide indoor navigation between rooms and floors, including stairs and elevators.
3. Support accessible navigation by avoiding stairs and preferring elevators and ramps.
4. Provide emergency navigation, including nearest-exit identification and alternative evacuation routes.
5. Implement and meaningfully use graphs, queues, priority queues/heaps, linked lists, stacks, and arrays/vectors. Provide graphical visualization of routes, graph traversal, and data-structure operations.
6. Support dynamic updates to buildings and pathways, including blocking and unblocking routes.
7. Maintain route history and provide backtracking using a stack.
8. Save and load campus data using file handling.
9. Provide dashboards, statistics, reports, input validation, and error handling.

Initial and Other Milestones:

Milestone 1: Design the system architecture, campus graph, data structures, and GUI structure.

Milestone 2: Implement campus locations, pathways, and core data structures.

Milestone 3: Implement BFS, DFS, and Dijkstra's algorithm with route reconstruction.

Milestone 4: Develop outdoor and indoor navigation, including floors, rooms, stairs, and elevators.

Milestone 5: Add accessible and emergency navigation features.

Milestone 6: Integrate visualization, route history, dynamic updates, file handling, dashboard, and reports.

Final Milestone: Perform testing, complexity analysis, algorithm comparison, error handling, and finalize the GUI and complete project demonstration.

Project Approach (3 pts)

(Describe the solution, platforms and technologies. Max 300 words).

The proposed Campus Navigation System will be developed in C++ with a graphical user interface. The campus will be represented as a graph, where buildings, rooms, laboratories and other locations are represented as nodes, while pathways are represented as edges containing distance, travel time and availability information.

The system will implement BFS, DFS and Dijkstra's algorithm for traversal, route finding and shortest-path navigation. Core data structures including Graph, Queue, Priority Queue/Heap, Linked List, Stack and arrays/vectors will be implemented in C++ and used for meaningful system functions. The system will provide indoor and outdoor navigation, accessible routes, emergency evacuation, alternative paths, route history and backtracking. It will also support dynamic updates such as adding or removing locations, modifying pathways, and blocking or unblocking routes.

The GUI will provide interactive map visualization, route highlighting, search, statistics, reports and step-by-step algorithm visualization to demonstrate the practical application of data structures and algorithms.

System Architecture (High Level Diagram)(2 pts)

(Provide an overview of the system and its main components in a diagram using a tool of your choice).

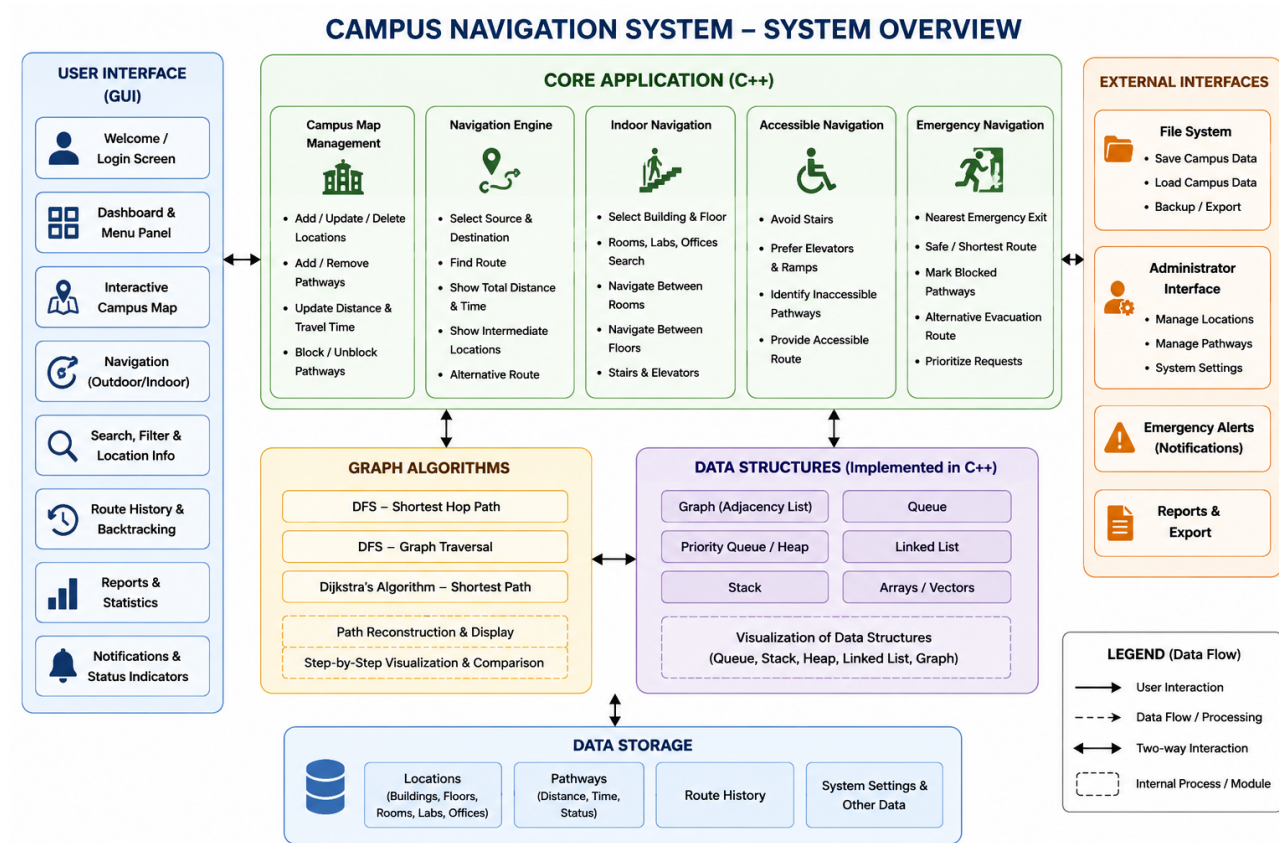


Figure 1: System Overview of the Campus Navigation System

Project Outcome / Deliverables (1 pts)

(Describe the project outcomes / deliverables. Max 200 words).

The primary outcome of the project will be a fully functional C++-based Campus Navigation System with an interactive graphical user interface. The system will provide a visual campus map and support navigation across buildings, floors, rooms, laboratories, offices, and other facilities. The major deliverables will include:

1. An interactive GUI with a welcome/login screen, dashboard, campus map, navigation controls, notifications, and validation messages.
2. A graph-based campus representation with dynamically manageable buildings, locations, and pathways.
3. Implementation of BFS, DFS, and Dijkstra's algorithm for graph traversal and route calculation, including path reconstruction and visualization.
4. Indoor and outdoor navigation with route distance, estimated travel time, and alternative routes.
5. Accessible navigation supporting stair avoidance and preference for elevators and ramps.
6. Emergency navigation for identifying emergency exits and generating alternative evacuation routes.
7. Implementation and visualization of required data structures including graphs, queues, priority queues/heaps, linked lists, stacks, and arrays/vectors.
8. Dynamic campus updates, route history and backtracking, search/filtering, file-based data storage, dashboard statistics, and reports.
9. Documentation covering time and space complexity, algorithm comparisons, and the suitability of selected data structures and algorithms.

The final deliverable will demonstrate the practical application of data structures and algorithms through a functional, interactive navigation system rather than a conventional management application.

Assumptions

(Describe the assumptions used to solve the problem. Max 100 words)

The system assumes that campus buildings, floors, rooms, laboratories, offices, facilities, and pathways can be represented as nodes and connections in a graph. Pathways are assumed to have known distances, estimated travel times, and availability status. It is assumed that users provide valid source and destination locations. The system assumes that accessibility information, such as stairs, elevators, ramps, and inaccessible pathways, is available. For emergency navigation, emergency exits and blocked pathways are assumed to be identified in the system. Campus data is assumed to be updated when locations or pathways change.

References

(Provide resources or references used for this deliverable).

1. cppreference — Reference for C++ language features, standard library, and programming concepts.
Link : <https://en.cppreference.com>
2. GeeksforGeeks – Data Structures and Algorithms — Reference for graphs, BFS, DFS, Dijkstra's algorithm, stacks, queues, and linked lists.
Link : <https://www.geeksforgeeks.org>
3. Programiz – Data Structures and Algorithms — Reference for fundamental data structures and graph algorithms.
Link : <https://www.programiz.com>
4. OpenStreetMap — Reference for understanding map-based representations of locations, pathways, and geographic navigation.
Link : <https://www.openstreetmap.org>